

Chemistry

Units 3 and 4

Practice Exam

This sample test has been prepared as part of the Pearson suite of resources for the Year 12, Units 3 and 4 ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes Working time: 180 minutes

Materials required

An approved non-programmable calculator.

Chemistry Data Booklet. This may be downloaded from the SCSA website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section 1: Multiple choice	25	25	50	25	14
Section 2: Short answer	12	12	60	74	40
Section 3: Extended answer	5	5	70	86	46
				Total	100

Section 1: Multiple choice

14% (25 marks)

This section has 25 questions. Answer **all** questions by circling the correct option. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 50 minutes

- 1 Consider the following statements about the conditions that are necessary for a reaction to occur.

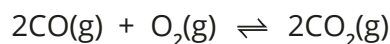
- I A collision must occur between reactant particles.
- II The energy of collision must be less than the activation energy.
- III Reactant particles must have the correct orientation.

Which of these statements are correct?

- A I and II only
- B I and III only
- C II and III only
- D I, II and III

The following information refers to Questions 2 and 3.

The reaction between carbon monoxide and oxygen is



where $K_c = 25$ at 400°C .

- 2 What is the equilibrium law expression for this reaction?

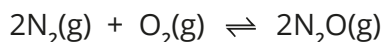
- A $\frac{2[\text{CO}_2]}{2[\text{CO}][\text{O}_2]}$
- B $\frac{2[\text{CO}][\text{O}_2]}{2[\text{CO}_2]}$
- C $\frac{[\text{CO}_2]^2}{[\text{CO}]^2[\text{O}_2]}$
- D $\frac{[\text{CO}]^2[\text{O}_2]}{[\text{CO}_2]^2}$

- 3 Which one of the following is the equation for the reaction that has $K_c = 0.04$? (The reaction also occurs at 400°C .)

- A $\text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightleftharpoons \text{CO}_2\text{(g)}$
- B $4\text{CO(g)} + 2\text{O}_2\text{(g)} \rightleftharpoons 4\text{CO}_2\text{(g)}$
- C $2\text{CO}_2\text{(g)} \rightleftharpoons 2\text{CO(g)} + \text{O}_2\text{(g)}$
- D $\text{CO}_2\text{(g)} \rightleftharpoons \text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)}$

The following information refers to Questions 4 and 5.

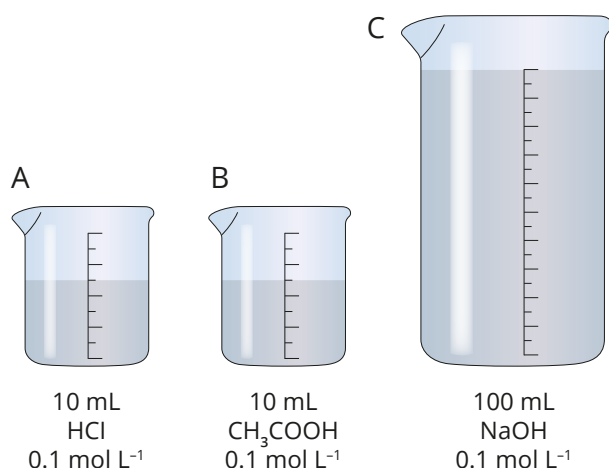
An equilibrium reaction occurs between nitrogen(I) oxide (N_2O) and nitrogen and oxygen gases:



At 20°C , $K_c = 1.3 \times 10^{-38}$ and at 50°C , $K_c = 3.1 \times 10^{-36}$.

- 4 Based on the data provided, which one of the following is a valid conclusion?
- A The reverse reaction is endothermic.
 - B The activation energy of the forward reaction is greater than the activation energy of the reverse reaction.
 - C At 20°C , the rate of the forward reaction in the equilibrium system is greater than the rate of the reverse reaction.
 - D The yield of N_2O is decreasing as the temperature increases.
- 5 A mixture of N_2O , N_2 and O_2 reaches equilibrium at 50°C . Which one of the following can be deduced from the information provided?
- A The concentration of O_2 in the equilibrium mixture must be half that of N_2 .
 - B The concentration of N_2 must be greater than that of O_2 .
 - C The equilibrium concentrations of all gases will be equal.
 - D The concentration of N_2O will be much less than that of O_2 .
- 6 Which statement about the following reaction is true?
- $$\text{NH}_4^+(\text{aq}) + \text{S}^{2-}(\text{aq}) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{HS}^-(\text{aq})$$
- A NH_4^+ acts as a base.
 - B S^{2-} acts as an acid.
 - C HS^- is the conjugate acid of NH_3 .
 - D NH_3 is the conjugate base of NH_4^+ .
- 7 Four beakers contain samples of acid as described below.
- Beaker I 1 mL of $0.0001 \text{ mol L}^{-1} \text{ HCl}$
- Beaker II 1 mL of $5.0 \text{ mol L}^{-1} \text{ HCl}$
- Beaker III 2500 mL of $0.01 \text{ mol L}^{-1} \text{ HCl}$
- Beaker IV 2500 mL of $5.0 \text{ mol L}^{-1} \text{ HCl}$
- Which beaker(s) contains the solution that has the highest pH?
- A I only
 - B II only
 - C I and III only
 - D II and IV only

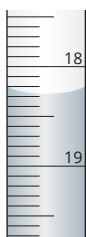
8 Three beakers are arranged with solutions as shown.



When 10 mL of water is added to each beaker, which beaker(s) experience the greatest change in pH?

- A beaker A only
 - B beaker B only
 - C beaker C only
 - D beakers A and B only
- 9 A beaker of pure water is cooled to 1°C. The concentration of hydroxide ion in this sample is found to be $1.0 \times 10^{-7.5}$ mol L⁻¹. Which one of the following statements about water at 1°C is true?
- A The water is alkaline.
 - B The pH of the water is 7.0.
 - C The value of K_w is 10^{-15} .
 - D The hydronium ion concentration is $1.0 \times 10^{-6.5}$ mol L⁻¹.
- 10 Select the correct statement about the use of indicators in acid–base titrations.
- A The acidity constant of the indicators must be approximately equal to the acidity constant of the acid.
 - B Acid–base indicators must change colour at pH 7 when the acid and base have neutralised each other.
 - C Phenolphthalein is a good indicator for all titrations because its colour change is so distinctive.
 - D An indicator should change colour at the point when the reactants are in the mole ratio as per the reaction equation.

11 A section of a burette filled with an acid is shown below.

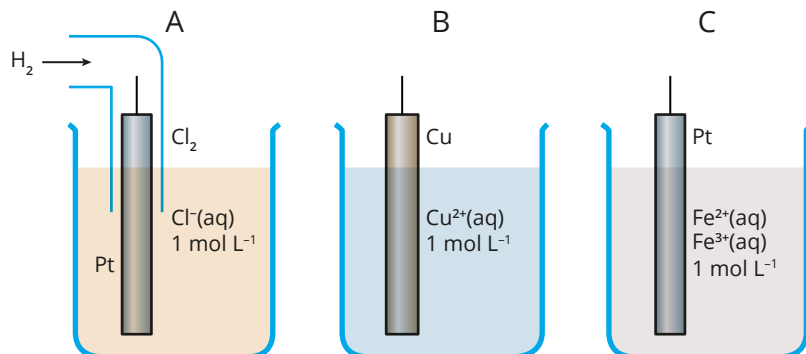


What is the correct reading on the burette scale?

- A 18.20 mL
 - B 18.25 mL
 - C 19.75 mL
 - D 19.80 mL
- 12 In which compound or ion does the element written in **bold** have the same oxidation number as chromium in Cr_2O_3 ?
- A AlCl_3
 - B NH_3
 - C SO_3^{2-}
 - D MnO_4^-

The following information refers to Questions 13 and 14.

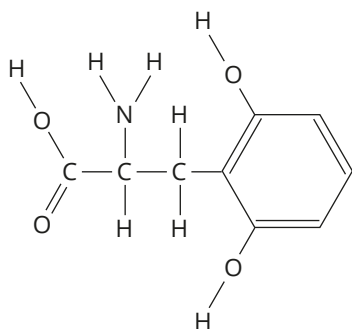
13 The three half-cells shown are available to form galvanic cells. The three half-cells are operating under standard conditions.



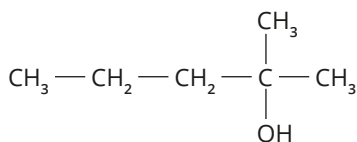
When cells B and C are connected, it is observed that the mass of the copper electrode decreases. What does this mean?

- A The copper ions are being reduced and the copper electrode will be the negative electrode.
 - B Oxidation is occurring at the anode and the anode will be the positive electrode.
 - C Fe^{2+} ions are being reduced and the platinum electrode will be the positive electrode.
 - D The copper atoms are being oxidised and the copper electrode will be the negative electrode.
- 14 Which one of the following is the correctly predicted potential difference produced when cells A and C are connected?
- A -0.69 V
 - B -0.18 V
 - C 0.59 V
 - D 2.13 V

- 15** Which one of the following reactions, carried out under standard conditions, is spontaneous?
- A** Iron(III) chloride solution is added to magnesium chloride solution.
 - B** Potassium bromide solution is added to zinc chloride solution.
 - C** Zinc metal is added to potassium bromide solution.
 - D** Zinc metal is added to iron(III) chloride solution.
- 16** Impure copper produced from the smelting of copper ores contains small amounts of impurities such as silver, gold, nickel and zinc. Which one of the statements about the electro-refining of copper is correct?
- A** The electrolytic cell consists of an impure copper anode attached to the negative terminal of an external power source and a pure copper cathode attached to the positive terminal.
 - B** Copper and impurities that are stronger reducing agents than copper are oxidised at the anode.
 - C** Impurities that are weaker reducing agents than copper are reduced at the anode.
 - D** Copper ions are the strongest oxidising agents present in the electrolyte and are the only metal ions reduced at the cathode.
- 17** The molecule shown below is known as Dopa. It is used in the treatment of Parkinson's disease.



- Which one of the following lists the functional groups in Dopa?
- A** hydroxyl, amide and ester
 - B** hydroxyl, amine and carboxyl
 - C** hydroxyl, amide and carboxyl
 - D** hydroxyl, carboxyl and amide
- 18** Consider the following structural formula.



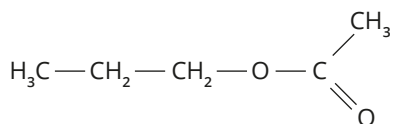
What is the systematic name of the compound represented by this structural formula?

- A** 4-methylpentan-4-ol
- B** 4-methylhexan-4-ol
- C** 2-methylpentan-2-ol
- D** 2-methylhexan-2-ol

19 Propene is reacted with chlorine gas. What is the molecular formula of the product formed?

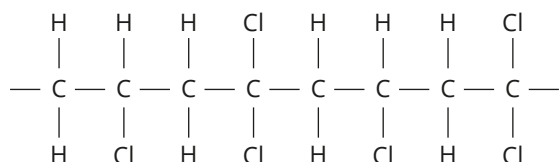
- A $C_3H_4Cl_2$
- B C_3H_5Cl
- C C_3H_7Cl
- D $C_3H_6Cl_2$

20 The molecule shown below could be formed from the reaction between which of the following?



- A propanoic acid and ethanol
- B ethanol and propanoic acid
- C propan-1-ol and methanoic acid
- D propan-1-ol and ethanoic (acetic) acid

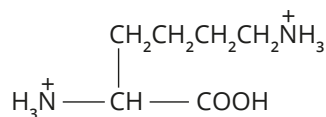
21 A section of a polymer containing two different monomers is shown below.



Which pair of monomers was used to synthesise this polymer?

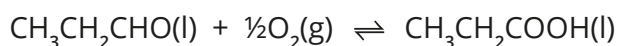
- A $\begin{array}{c} H & H \\ | & | \\ H-C & -C-H \\ | & | \\ H & Cl \end{array}$ and $\begin{array}{c} H & Cl \\ | & | \\ H-C & -C-H \\ | & | \\ H & Cl \end{array}$
- B $\begin{array}{c} H & H \\ | & | \\ HO-C & -C-OH \\ | & | \\ H & Cl \end{array}$ and $\begin{array}{c} H & H \\ | & | \\ HO-C & -C-OH \\ | & | \\ H & Cl \end{array}$
- C $\begin{array}{c} H & & H \\ & \diagdown & / \\ & C=C & \\ & / & \diagdown \\ H & & Cl \end{array}$ and $\begin{array}{c} H & & Cl \\ & \diagdown & / \\ & C=C & \\ & / & \diagdown \\ H & & Cl \end{array}$
- D $\begin{array}{c} H & & H \\ & \diagdown & / \\ & C=C & \\ & / & \diagdown \\ H & & H \end{array}$ and $\begin{array}{c} H & & Cl \\ & \diagdown & / \\ & C=C & \\ & / & \diagdown \\ Cl & & Cl \end{array}$

22 The following structural formula represents lysine at which pH value?



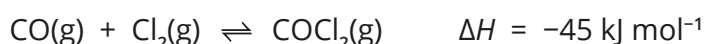
- A pH 12
- B pH 10
- C pH 7
- D pH 2

- 23** Cobalt (Co) is used as a catalyst in the industrial preparation of propanoic acid from propanal ($\text{CH}_3\text{CH}_2\text{CHO}$):

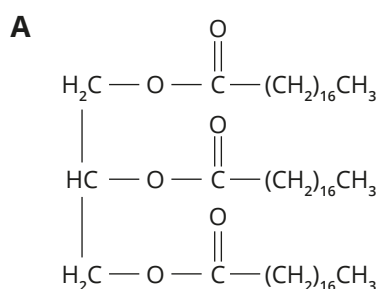


Which one of the following statements about the use of cobalt as a catalyst in this reaction is correct?

- A** Cobalt will not catalyse the reverse reaction.
 - B** Cobalt will improve the equilibrium yield at any given temperature.
 - C** Cobalt will be more effective in powdered form than granule form.
 - D** Cobalt will lower the enthalpy of the reaction.
- 24** Industrial chemists always seek to improve the yield of product in reversible reactions. For the following reaction, which alternative contains only changes that will maximise the equilibrium yield?



- A** Decreased temperature, excess of chlorine, high pressure
 - B** Increased temperature, use of a catalyst, addition of an inert gas
 - C** Increased temperature, excess of chlorine, high pressure
 - D** Excess of chlorine, low pressure, use of a catalyst
- 25** Which one of the following substances is soluble in soft water but forms a precipitate in hard water?



- B** $\text{CH}_3(\text{CH}_2)_{16}\text{COO}^-\text{Na}^+$
- C** $\text{CH}_3(\text{CH}_2)_{11}-\text{C}_6\text{H}_5$
- D** $\text{CH}_3(\text{CH}_2)_{11}-\text{C}_6\text{H}_4-\text{SO}_3^-\text{Na}^+$

End of section 1

Section 2: Short answer

40% (74 marks)

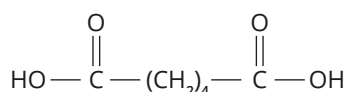
This section has 12 questions. Answer **all** questions. Write your answers in the spaces provided. When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable. Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them. Suggested working time: 60 minutes

Question 26

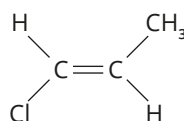
(10 marks)

The structures of a number of molecules that are able to undergo polymerisation reactions are provided below.

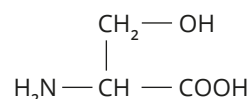
I



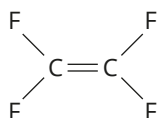
II



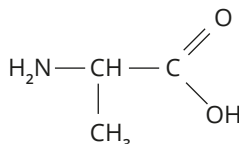
III



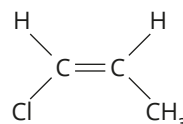
IV



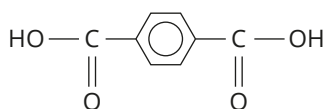
V



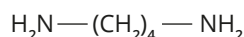
VI



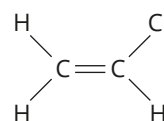
VII



VIII



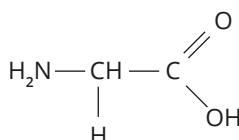
IX



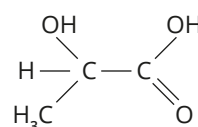
X



XI



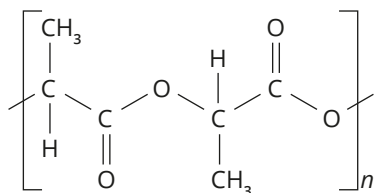
XII



Refer to the Roman numerals corresponding to these structures when answering the following questions.

- a Which of the monomers shown above are able to form addition polymers? (1 mark)
-
- b Identify the monomer used to manufacture the polymer that is used to provide the non-stick surfaces of cookware. (1 mark)
-
- c Which three monomers could be used to synthesise proteins? (1 mark)
-
- d Identify the two molecular structures that are *cis-trans* isomers. (1 mark)
-
- e Name the class of condensation polymer formed by the reaction between compound I and compound VIII. (1 mark)
-

f The structure of a small section of a polymer is shown below.



i Which monomer was used to synthesise this polymer? (1 mark)

ii Is the reaction that formed this polymer an addition or condensation polymerisation reaction? Justify your answer. (2 marks)

g Draw the structure of a peptide that can be formed from a reaction between three different molecules represented on the previous page. (2 marks)

Question 27

(10 marks)

a Draw the structural formulae of a primary alcohol and a secondary alcohol that have the molecular formula C_3H_8O . (2 marks)

Primary alcohol	Secondary alcohol

b Propanoic acid is one of two possible products that can be formed when the primary alcohol reacts with an acidified potassium dichromate solution. Provide the name and structural formula of the other product. To what category of compounds does this product belong? (3 marks)

Name	Structural formula
Category	

- c** The secondary alcohol that has the molecular formula C_3H_8O is also oxidised by acidified dichromate solution. Identify the category of compounds to which the product of this reaction belongs. Name the product and draw its structural formula. (3 marks)

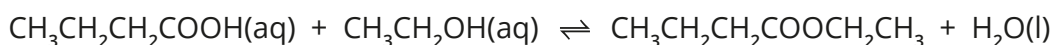
Name	Structural formula
Category	

- d** The dichromate ion is reduced in the reactions with primary and secondary alcohols.
- i** What colour change would you expect to observe as the dichromate ion is reduced? (1 mark)
- _____
- ii** Write the half equation for the oxidation of the primary alcohol to form the carboxylic acid. (1 mark)
- _____
- _____

Question 28

(6 marks)

Ethanol and butanoic acid react to form a product that is used as a flavour enhancer in some fruit juices and alcoholic drinks:



- a** Give the name of the organic compound that is produced in this reaction. (1 mark)
- _____
- b** In a synthesis, 0.8811 g of butanoic acid was reacted with 0.7410 g of ethanol. The reaction products were separated by distillation and 0.907 g of the flavouring agent was obtained. The molar masses the compounds involved in this reaction are:
- butanoic acid 88.11 g mol⁻¹
 - ethanol 46.07 g mol⁻¹
 - flavour enhancer 116.2 g mol⁻¹.
- i** Identify the limiting reagent. (2 marks)
- _____
- _____
- ii** What is the percentage yield for this reaction? (3 mark)
- _____
- _____
- _____
- _____

Question 29**(4 marks)**

Both the ammonium and carbonate ions undergo hydrolysis reactions when ammonium carbonate is dissolved in water.

a i Write a balanced equation for the hydrolysis of the ammonium ion. (1 mark)

ii Write a balanced equation for the hydrolysis of the carbonate ion. (1 mark)

iii Name the components of one acid–base conjugate pair in the reaction represented by equation ii. (1 mark)

b Explain why a solution of ammonium carbonate has a pH greater than 7 at 25°C. (1 mark)

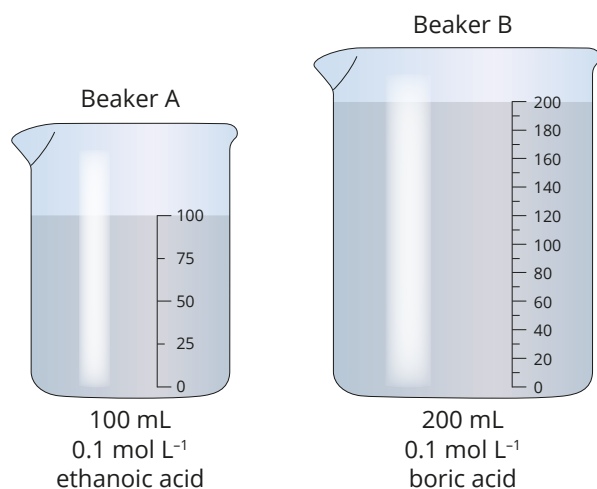
Question 30**(3 marks)**

a Draw the molecular structure of methanoic acid. (1 mark)

b Is methanoic acid a monoprotic or diprotic acid? Justify your answer. (2 marks)

Question 31**(4 marks)**

Beaker A contains 100 mL of 0.1 mol L⁻¹ ethanoic acid. Beaker B contains 200 mL of 0.1 mol L⁻¹ hypochlorous acid (HOCl).



- a** The concentration of the hydronium ions in beaker A is 1.34×10^{-3} mol L⁻¹. Calculate the pH of the solution. (1 mark)

- b** The acidity constant of ethanoic acid is 1.5×10^{-5} and the acidity constant of hypochlorous acid is 3.5×10^{-8} .

- i** Which beaker has the higher pH? (1 mark)

- ii** Which beaker has the greater amount of hydronium ions? (1 mark)

- iii** Which beaker has the higher electrical conductivity? (1 mark)

Question 32**(6 marks)**

A buffer solution is prepared by mixing 100 mL of 0.1 mol L⁻¹ hydrofluoric acid (HF) solution with 100 mL of a 0.1 mol L⁻¹ sodium fluoride solution. The acidity constant of HF is 6.6×10^{-4} .

- a** Write an equation for the ionisation of HF in water. (1 mark)

- b** Write the equilibrium law expression for the acidity constant of HF. (1 mark)

- c Explain, with the aid of appropriate equations, why the pH of this buffer solution does not change greatly on the addition of a small amount of a:

- strong base

- strong acid

(4 marks)

Question 33

(6 marks)

Hydrogen peroxide (H_2O_2) is listed in two separate half equations in the electrochemical series in the table of Standard Reduction Potentials at 25°C provided in the Data Book.

- a Write a balanced overall equation for the decomposition of hydrogen peroxide. (1 mark)

- b The electrochemical data suggests that hydrogen peroxide will react spontaneously. However, it can be stored in a dark container for several years. Explain why this is possible. (1 mark)

- c Complete the table below by identifying a reactant that will:

- oxidise hydrogen peroxide
- reduce hydrogen peroxide.

Write a full and balanced equation for each reaction.

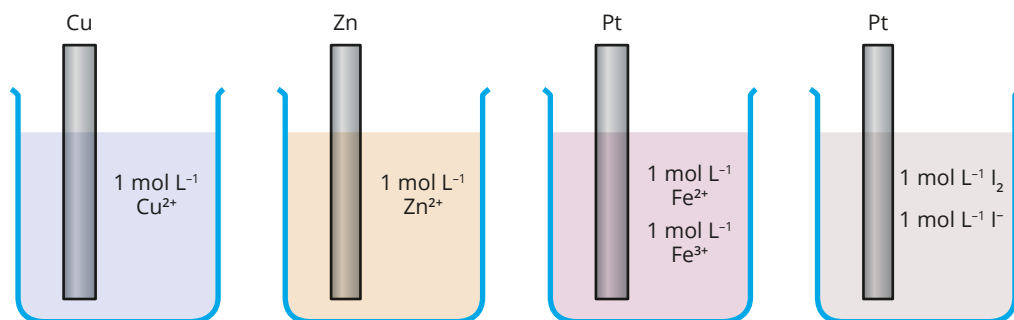
Use the information provided in the table of Standard Reduction Potentials at 25°C in the Data Book to help you answer this question. (4 marks)

H_2O_2 is ...	by reactant	Equation
oxidised		
reduced		

Question 34

(8 marks)

You have been provided with the following half-cells.



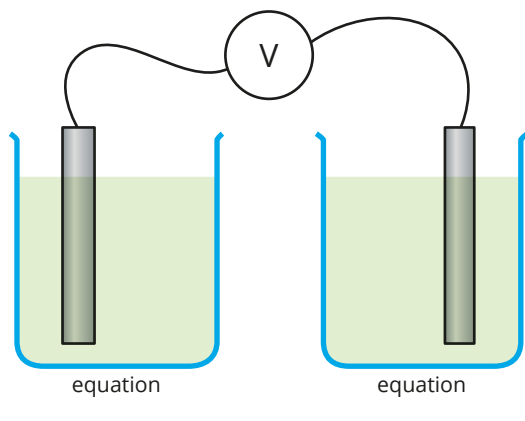
A voltmeter, wires with clips, filter paper and a saturated solution of potassium nitrate are also provided.

- a i** Which half-cells can be combined to form a galvanic cell that produces the highest voltage at standard conditions? (2 marks)

- ii** What is the greatest voltage, at standard conditions, that can be produced by this combination of these half-cells? (1 mark)

- b** A partly completed cell diagram for this cell is provided below. Complete and label the diagram. You should also:

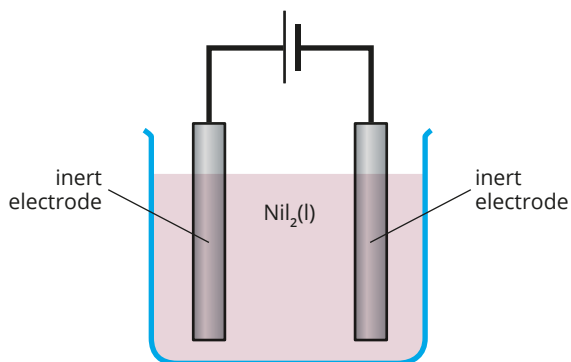
- show the direction of current flow through the wires
- label the polarity of each electrode
- provide an equation for each half-cell reaction.



(5 marks)

Question 35**(6 marks)**

Molten nickel iodide (NiI_2) is subjected to electrolysis with inert electrodes.



- a** Use an arrow to indicate the direction of electron flow. (1 mark)
- b** Label the anode and the cathode. (2 marks)
- c i** Write half-equations for the reactions occurring at each electrode.
anode: _____ (1 mark)
cathode: _____ (1 mark)
- ii** Write an overall equation for the reaction occurring in this cell. (1 mark)

Question 36**(5 marks)**

Proteins are described as having three structural levels: primary, secondary and tertiary structures. Briefly discuss the nature of the bonding or interactions that are responsible for each structural level.

Structure	Bonding and interactions
primary	
secondary	
tertiary	

Question 37**(6 marks)**

Ethanol is produced on an industrial scale by the:

- catalysed hydrolyses of ethene derived from crude oil
- enzyme-catalysed fermentation of glucose derived from plants.

- a i** Write an equation for the production of ethanol from ethene. (1 mark)

- ii** Write an equation for the production of ethanol from the fermentation of glucos, ($\text{C}_6\text{H}_{12}\text{O}_6$). (1 mark)

- b** Both processes use catalysts to increase the rate of ethanol production. Briefly explain why catalysts increase reaction rates. (2 marks)

- c** Provide one economic advantage and one environmental advantage for using enzymes rather than inorganic catalysts to produce ethanol on an industrial scale.

- i** economic advantage (1 mark)

- ii** environmental advantage (1 mark)

End of section 2

Section 3: Extended answer

46% (86 marks)

This section contains 5 questions. You must answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 70 minutes

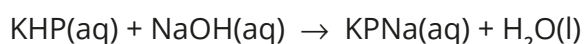
Question 38

(14 marks)

Dilute solutions of sodium hydroxide are sometimes used to determine the amount of acid in a solution. However, aqueous solutions of sodium hydroxide absorb carbon dioxide from the atmosphere, lowering the concentration of hydroxide ions in solution. Therefore, the exact concentration of hydroxide ion in a solution of sodium hydroxide needs to be determined before the solution is used in volumetric analysis.

The concentration of a solution of sodium hydroxide can be determined by titration against a solution of a primary standard such as the weak acid, potassium hydrogen phthalate ($\text{C}_8\text{H}_5\text{O}_4\text{K}$).

This reaction is represented by the following equation. For the sake of simplicity, potassium hydrogen phthalate is represented as KHP.



KHP is a weak acid. A table of pH ranges of a number of indicators is provided below.

Indicator	Colour of acid form	Colour of base form	pH range over which the colour changes
bromothymol blue	yellow	blue	6.0–7.6
methyl orange	pink	yellow	3.1–4.4
phenolphthalein	colourless	pink	8.3–10

- a** Distinguish between the end point and the equivalence point of a titration. (2 marks)

- b i** Which indicator would be best suited to detect the end point of this titration? Justify your selection. (2 marks)

- ii** If this indicator were to be used, what colour change would be observed during this titration? (1 mark)

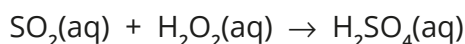
- c** Name one criterion that might be used to determine the suitability of KHP as a primary standard. (1 mark)

A mass of 0.04814 g of KHP was dissolved in a conical flask containing 50 mL of distilled water that had recently been boiled. This solution was titrated against the sodium hydroxide solution. A titre of 23.62 mL of was required.

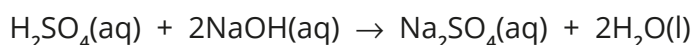
- d** Calculate the concentration of the sodium hydroxide solution. The molar mass of KHP is 204.2 g mol⁻¹. (3 marks)

The standardised NaOH solution is then used to determine the amount of sulfur dioxide in a sample of wine. Sulfur dioxide is added to wine as an antioxidant and to prevent the growth microorganism.

Hydrogen peroxide was added to a 100.0 mL sample of wine. The sulfur dioxide in the sample reacted as follows:



The solution of sulfuric acid was then titrated against the standardised solution of sodium hydroxide from part **a** of this question:



A titre of 13.44 mL of the standardised NaOH solution was required.

- e** Calculate the sulfur dioxide content of the wine in mg L⁻¹. (5 marks)

(24 marks)

a What is the empirical formula of this compound? (10 marks)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

b Determine the molecular formula of compound X. (4 marks)

c Draw the two possible structural formula for compound X and provide the IUPAC name that corresponds to each structure. (4 marks)

	Compound I	Compound II
Structural formula		
Name		

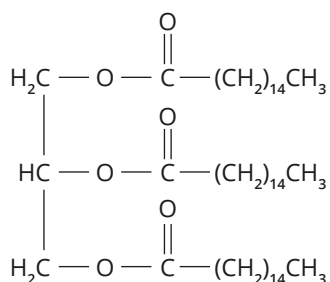
- d** Describe two tests that can be used to differentiate between the two possible identities of compound X. Briefly describe the results you would expect for each test for the two possible identities. (6 marks)

Description of test	Test results for compound I	Test results for compound II
Test 1		
Test 2		

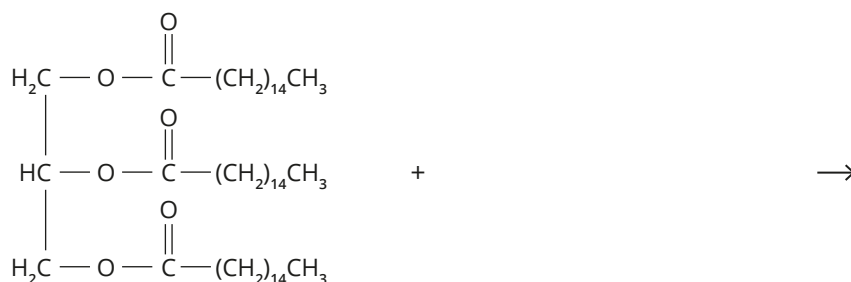
Question 40

(16 marks)

The triglyceride, glycerol tripalmitate, is obtained from palm oil and is used in the manufacture of soap. The structure of this triglyceride is provided below.



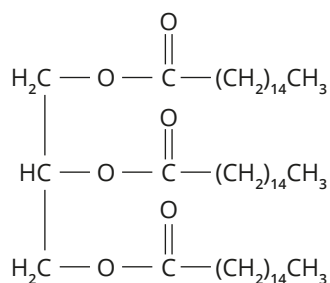
- a** Complete the equation for this saponification reaction. (2 marks)



- b** Briefly explain the cleaning action of soap derived from glycerol tripalmitate in terms of its structure and the nature of its interaction between water molecules and grease particles. You may use diagrams to support your explanation. (6 marks)

- c Explain why soaps form a scum in hard water but detergents do not form a scum. Include an appropriate equation in your explanation. (3 marks)

- d Base catalysis is also used to convert palm oil to biodiesel. Complete the equation for the production of biodiesel from glycerol tripalmitate using the base catalysed method. (2 marks)

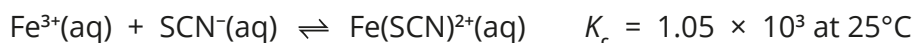


- e Briefly distinguish between the base-catalysed and lipase-catalysed methods used to convert glycerol tripalmitate to biodiesel. Identify one advantage of using the lipase-catalysed method over the base-catalysed method. (3 marks)

Question 41

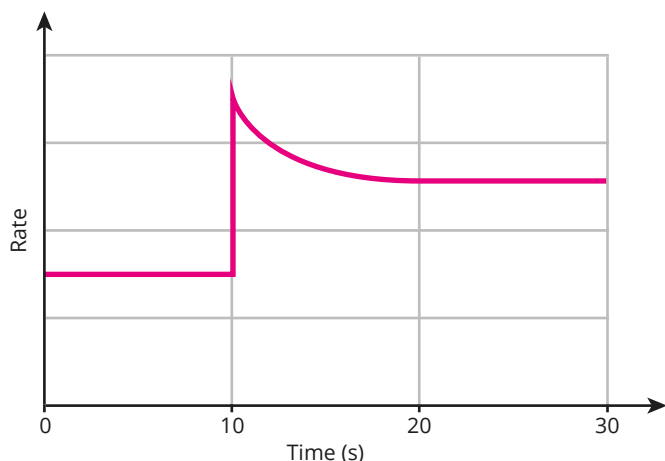
(16 marks)

The following equilibrium is established when solutions of $0.1 \text{ mol L}^{-1} \text{Fe(NO}_3)_2$ and $0.1 \text{ mol L}^{-1} \text{NaSCN}$ are mixed:



The Fe(SCN)^{2+} ion has a red colour, the intensity of which depends on its concentration.

- a** The rate of the forward reaction is measured before, during and after the addition of a small amount of solid $\text{Fe(NO}_3)_3$ to a mixture of Fe^{3+} , SCN^{-} and Fe(SCN)^{2+} that was initially at equilibrium. The change in rate is shown in the following graph.



- i** Explain, in terms of collision theory, the changes to the equilibrium system:

- between 0 and 10 seconds
- at 10 seconds
- between 10 and 20 seconds
- between 20 and 30 seconds.

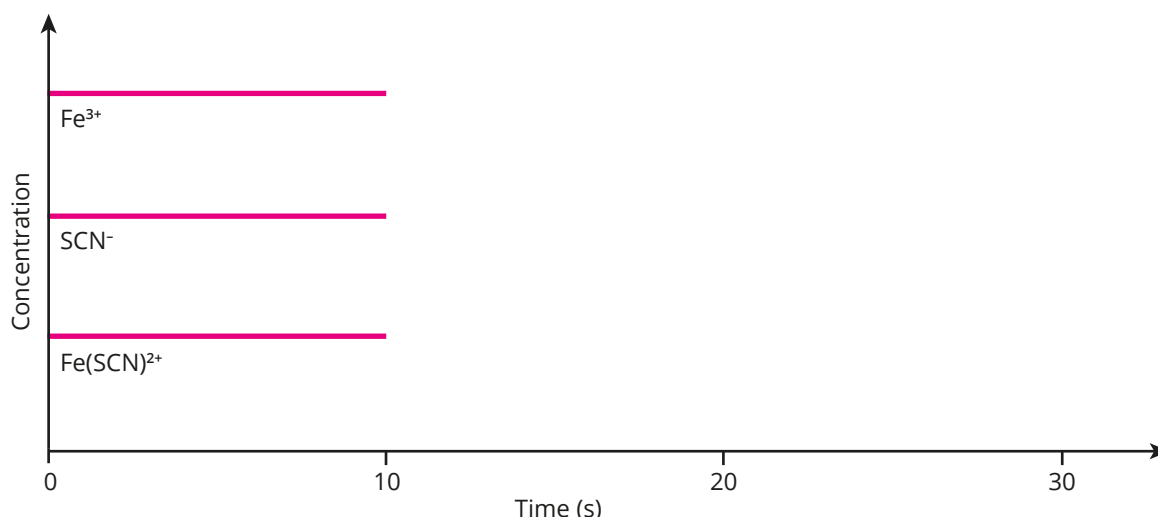
(4 marks)

Time	Explanation
between 0 and 10 seconds	
at 10 seconds	
between 10 and 20 seconds	
between 20 and 30 seconds	

- ii** Explain why the rate of the forward reaction is greater between 20 and 30 seconds than the rate between 0 and 10 seconds. (1 mark)

- b** On the graph above, show how the rate of the reverse reaction changes between 0 and 30 seconds. (3 marks)

- c The concentrations of Fe^{3+} , SCN^- and $\text{Fe}(\text{SCN})^{2+}$ between 0 and 10 seconds are plotted on the following graph. On the graph, show the changes in concentration for each of these ions between 10 and 30 seconds. (3 marks)



- d How could you determine experimentally whether the forward reaction is exothermic or endothermic? What macroscopic property of the reaction system would you measure? How would you expect this macroscopic property to change if the reaction is exothermic or endothermic? Justify your predictions in terms of Le Châtelier's principle. (5 marks)

Question 42

(16 marks)

The reaction occurring in the converter in the contact process is:



- a A number of changes are made to the reaction systems, as indicated in the table below.
- i Insert the terms *increase*, *decrease* and *no change* to describe the effect of each change on the yield and rate of production of SO_3 . (6 marks)

Change	Effect on yield of SO_2	Effect on rate of production of SO_3
increase in temperature at constant volume		
decrease in pressure at constant temperature		
addition of a catalyst at constant temperature and volume		

- ii** Predict, with reasoning, using Le Châtelier's principle, the effect of decreasing the pressure on the yield of SO_3 . (3 marks)

- iii** Predict the effect of a temperature increase on the equilibrium system. Use the concept of reaction rates to explain this effect. (3 marks)

- b** Consider the following statement.

The industrial scale synthesis of some chemicals involves a compromise between the reaction conditions that lead to maximum yield and conditions that lead to maximum rate of production.

Discuss how this statement is applicable to the production of SO_3 . (4 marks)

End of questions